

OOPS ASSIGNMENT

Question1.

Code :

```
#include<iostream>
```

```
using namespace std;
```

```
template<typename T>
```

```
void swapValues(T &a, T &b) {
```

```
    T temp = a;
```

```
    a = b;
```

```
    b = temp;
```

```
}
```

```
int main() {
```

```
    int x = 10, y = 20;
```

```
    cout << "Before swap: x = " << x << ", y = " << y << endl;
```

```
    swapValues(x, y);
```

```
    cout << "After swap: x = " << x << ", y = " << y << endl;
```

```
    double p = 3.14, q = 2.71;
```

```
    cout << "Before swap: p = " << p << ", q = " << q << endl;
```

```
    swapValues(p, q);
```

```
    cout << "After swap: p = " << p << ", q = " << q << endl;
```

```
    char c1 = 'A', c2 = 'Z';
```

```
    cout << "Before swap: c1 = " << c1 << ", c2 = " << c2 << endl;
```

```
    swapValues(c1, c2);
```

```
    cout << "After swap: c1 = " << c1 << ", c2 = " << c2 << endl;
```

```
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Before swap: x = 10, y = 20
After swap: x = 20, y = 10
Before swap: p = 3.14, q = 2.71
After swap: p = 2.71, q = 3.14
Before swap: c1 = A, c2 = Z
After swap: c1 = Z, c2 = A
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> █
```

Question2.

```
#include<iostream>
```

```
using namespace std;
```

```
template<typename T>
```

```
T findMin(T arr[], int n) {
```

```
    T minVal = arr[0];
```

```
    for(int i = 1; i < n; i++)
```

```
        if(arr[i] < minVal) minVal = arr[i];
```

```
    return minVal;
```

```
}
```

```
int main() {
```

```
    int intArr[] = {5, 2, 8, 1, 9};
```

```
    cout << "Min in int array: " << findMin(intArr, 5) << endl;
```

```
    double dblArr[] = {3.5, 1.2, 7.8, 0.9};
```

```
    cout << "Min in double array: " << findMin(dblArr, 4) << endl;
```

```
    char charArr[] = {'d', 'a', 'z', 'b'};
```

```
    cout << "Min in char array: " << findMin(charArr, 4) << endl;
```

```
    return 0;
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Min in int array: 1
Min in double array: 0.9
Min in char array: a
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question3.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
template<typename T>
```

```
void bubbleSort(T arr[], int n) {
```

```
    for(int i = 0; i < n-1; i++)
```

```
        for(int j = 0; j < n-i-1; j++)
```

```
            if(arr[j] > arr[j+1]) swap(arr[j], arr[j+1]);
```

```
}
```

```
int main() {
```

```
    int arr[] = {64, 25, 12, 22, 11};
```

```
    int n = 5;
```

```
    cout << "Before sort: ";
```

```
    for(int i=0; i<n; i++) cout << arr[i] << " ";
```

```
    bubbleSort(arr, n);
```

```
    cout << "\nAfter sort: ";
```

```
    for(int i=0; i<n; i++) cout << arr[i] << " ";
```

```
    cout << endl;
```

```
    double darr[] = {3.5, 1.2, 7.8, 0.9};
```

```
    int m = 4;
```

```
    cout << "Before sort: ";
```

```
for(int i=0; i<m; i++) cout << darr[i] << " ";  
bubbleSort(darr, m);  
cout << "\nAfter sort: ";  
for(int i=0; i<m; i++) cout << darr[i] << " ";  
cout << endl;  
return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }  
Before sort: 64 25 12 22 11  
After sort: 11 12 22 25 64  
Before sort: 3.5 1.2 7.8 0.9  
After sort: 0.9 1.2 3.5 7.8  
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question4.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
template<typename T>
```

```
int linearSearch(T arr[], int n, T key) {
```

```
    for(int i = 0; i < n; i++)
```

```
        if(arr[i] == key) return i;
```

```
    return -1;
```

```
}
```

```
int main() {
```

```
    int arr[] = {10, 25, 30, 45, 60};
```

```
    int key = 30;
```

```
    int result = linearSearch(arr, 5, key);
```

```
    if(result != -1)
```

```
        cout << "Element " << key << " found at index " << result << endl;
```

```
    else
```

```
        cout << "Element not found" << endl;

double darr[] = {1.1, 2.2, 3.3, 4.4};
double dkey = 3.3;

int res2 = linearSearch(darr, 4, dkey);
if(res2 != -1)
    cout << "Element " << dkey << " found at index " << res2 << endl;
else
    cout << "Element not found" << endl;
return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Element 3.3 found at index 2
Element 3.3 found at index 2
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question5.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
// i) Single parameter
```

```
template<typename T>
```

```
void process(T a) {
```

```
    cout << "Single parameter: " << a << endl;
```

```
}
```

```
// ii) Two parameters of same type
```

```
template<typename T>
```

```
void process(T a, T b) {
```

```
    cout << "Two parameters (same type): " << a << " and " << b << endl;
```

```
}

// iii) Two parameters of different types
template<typename T1, typename T2>
void process(T1 a, T2 b) {
    cout << "Two parameters (different types): " << a << " and " << b << endl;
}

int main() {
    process(42);
    process(3.14);
    process(10, 20);
    process(1.5, 2.5);
    process(100, 3.14);
    process('A', 99);
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ tempCodeRunnerFile.CPP -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Single parameter: 42
Single parameter: 3.14
Two parameters (same type): 10 and 20
Two parameters (same type): 1.5 and 2.5
Two parameters (different types): 100 and 3.14
Two parameters (different types): A and 99
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question6.

Code:

```
for(int i=0; i<=top; i++) cout << arr[i] << " ";

cout << endl;

}

};

int main() {
```

```
Stack<int> s;  
  
s.push(10); s.push(20); s.push(30);  
  
s.display();  
  
s.pop();  
  
s.display();  
  
return 0;  
  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "C:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }  
10 pushed to stack  
20 pushed to stack  
30 pushed to stack  
Stack: 10 20 30  
30 popped from stack  
Stack: 10 20  
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question7.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
template<typename T>
```

```
class Queue {
```

```
    T arr[100];
```

```
    int front, rear;
```

```
public:
```

```
    Queue() { front = rear = -1; }
```

```
    void enqueue(T val) {
```

```
        if(rear == 99) { cout << "Queue is full!" << endl; return; }
```

```
        if(front == -1) front = 0;
```

```
        arr[++rear] = val;
```

```
        cout << val << " enqueued" << endl;
```

```
    }
```

```
    void dequeue() {
```

```
        if(front == -1 || front > rear) { cout << "Queue is empty!" << endl; return; }

        cout << arr[front++] << " dequeued" << endl;
    }

    void display() {
        if(front == -1 || front > rear) { cout << "Queue is empty" << endl; return; }

        cout << "Queue: ";
        for(int i=front; i<=rear; i++) cout << arr[i] << " ";
        cout << endl;
    }
};

int main() {
    Queue<int> q;
    q.enqueue(10); q.enqueue(20); q.enqueue(30);
    q.display();
    q.dequeue();
    q.display();
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
10 enqueued
20 enqueued
30 enqueued
Queue: 10 20 30
10 dequeued
Queue: 20 30
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question8.

Code:

```
#include<iostream>

#include<string>

using namespace std;
```



```
template<typename T1, typename T2>
class Pair {
    T1 first;
    T2 second;
public:
    Pair(T1 a, T2 b) : first(a), second(b) {}
    void display() {
        cout << "Pair: (" << first << ", " << second << ")" << endl;
    }
};
```

```
int main() {
    Pair<int, int> p1(10, 20);
    p1.display();

    Pair<string, int> p2("Age", 21);
    p2.display();

    Pair<double, char> p3(3.14, 'A');
    p3.display();
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Pair: (10, 20)
Pair: (Age, 21)
Pair: (3.14, A)
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question9.

Code:

```
#include<iostream>

using namespace std;
```

```
template<typename T>
class Arithmetic {
    T a, b;
public:
    Arithmetic(T x, T y) : a(x), b(y) {}
    T add()    { return a + b; }
    T subtract() { return a - b; }
    T multiply() { return a * b; }
    T divide() { return (b != 0) ? a / b : 0; }
    void display() {
        cout << "Add:    " << add()    << endl;
        cout << "Subtract: " << subtract() << endl;
        cout << "Multiply: " << multiply() << endl;
        cout << "Divide:  " << divide() << endl;
    }
};

int main() {

    Arithmetic<int> obj1(20, 4);
    cout << "Integer Arithmetic:" << endl;
    obj1.display();

    Arithmetic<double> obj2(7.5, 2.5);
    cout << "Double Arithmetic:" << endl;
    obj2.display();
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "C:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Integer Arithmetic:
Add:      24
Subtract: 16
Multiply: 80
Divide:   5
Double Arithmetic:
Add:      10
Subtract: 5
Multiply: 18.75
Divide:   3
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question10.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
template<typename T>
```

```
class ArrayTemplate {
```

```
    T arr[100];
```

```
    int n;
```

```
public:
```

```
    void input() {
```

```
        cout << "Enter number of elements: ";
```

```
        cin >> n;
```

```
        cout << "Enter elements:" << endl;
```

```
        for(int i=0; i<n; i++) cin >> arr[i];
```

```
    }
```

```
    void display() {
```

```
        cout << "Array elements: ";
```

```
        for(int i=0; i<n; i++) cout << arr[i] << " ";
```

```
        cout << endl;
```

```
    }
```

```
};
```

```
int main() {
```

```
    ArrayTemplate<int> a;
```

```
    cout << "Integer Array:" << endl;
```

```
    a.input();
```

```
a.display();

ArrayTemplate<double> b;

cout << "Double Array:" << endl;

b.input();

b.display();

return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Integer Array:
Enter number of elements: 5
Enter elements:
4 5 6 3 4
Array elements: 4 5 6 3 4
Double Array:
Enter number of elements: 3
Enter elements:
3.3 5.3 2.3
Array elements: 3.3 5.3 2.3
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> 
```